Module-V

Lecture-III

Equivalences between CTL Formulas

1. **Introduction**

In the previous lecture we have discussed the syntax and semantics of CTL. Using these concepts, we also presented how logical statements are represented using well-formed CTL formulas. Further, in a given model, states satisfy a given CTL formula was illustrated using examples.

We know that for a given propositional logic formula, there are several equivalent formulas, for example, (i) $p \rightarrow q \equiv \neg p \vee q$, (ii) $p \; AND \; q \equiv \overline{\overline{p} \text{ OR } \overline{q}}$ etc. Similarly, in case of predicate logic also, a given formula can be represented in term of other equivalent ones, for example, $\neg \forall x(P(x)) \equiv \exists x(\neg P(x))$; $\neg \forall x(P(x))$: It is not the case that for all values of *x, P(x)* is true and $\exists x(\neg P(x))$: there exist *x* for which *P(x)* is not true.

Two logic formulas φ and Ψ are said to be semantically equivalent if for every truth assignment, both the formulas have the same values. Table 1 illustrates how $p \rightarrow q$ and $\neg p \vee q$ are semantically equivalent by assigning all possible truth values to p and q and verifying that both the formulas produce same values.

p	**q**	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Table 1. Semantic equivalence of $p \rightarrow q$ and $\neg p \vee q$

Similarly, in case of CLT one formula can be represented in terms of other equivalent ones. Two CTL formulas φ and Ψ are said to be semantically equivalent if in any state in any model which satisfies one of them also satisfies the other.; we denote this by $\varphi \equiv \Psi$.

Equivalence in CTL formulas is analogous to equivalence in propositional or predicate logic. In this lecture we will discuss equivalences of different CTL formulas.

2. **Equivalence of basic CTL formulas**

As already discussed in last lecture,

Quantifier: $A \equiv$ All path, $E \equiv$There exist a path.

Operator: $G \equiv$Global, $F \equiv$In future

Path Formula: $\boldsymbol{G\,\phi} \equiv$ Globally ϕ is true; in a path in every state ϕ is true.

$\boldsymbol{F\,\phi} \equiv$ In future ϕ is true; in a path there are some states in which ϕ is true.

$\boldsymbol{X\,\phi} \equiv$ in next state ϕ is true.

$\boldsymbol{\phi 1\ U\ \phi 2} \equiv \phi 1$ is true until $\phi 2$ becomes true.

Figure 1 illustrates examples where the above formulas hold.

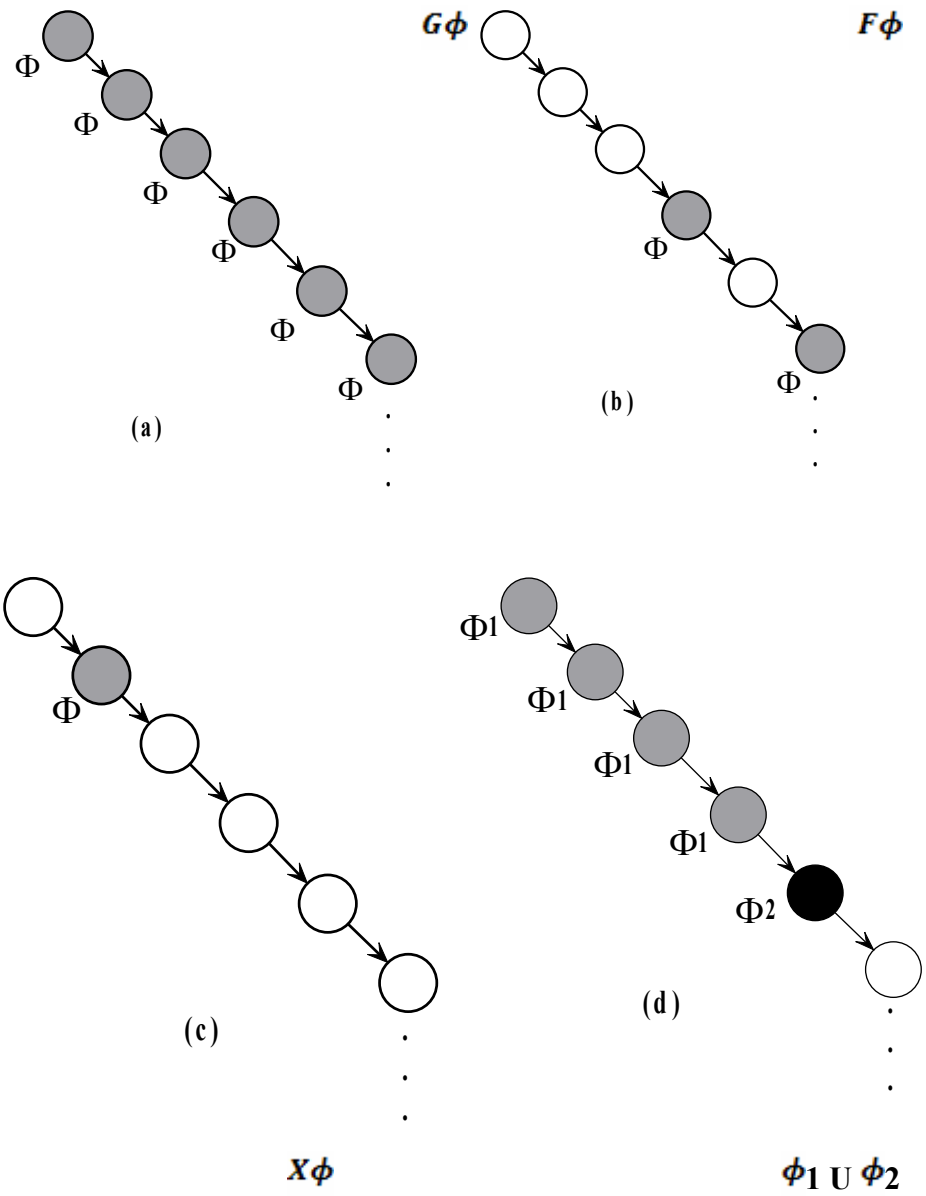


Figure 1. Illustration of path formulas

State formulas:

$\boldsymbol{AG\phi}$**:** From a given state, In all paths in all states $\boldsymbol{\phi}$ is true.

$EG\phi$: From a given state, There exists a path where in all the states ϕ is true.

$AF\phi$: From a given state, In all paths in future ϕ is true

$EF\phi$: From a given state, There exist a path where in future ϕ is true

$AX\phi$: From a given state, In all paths in the next state ϕ holds

$EX\phi$: From a given state, There exists a path in which in the next state ϕ holds.

A ($\phi 1\ U\ \phi 2$): From a given state, In all paths $\phi 1$ holds until $\phi 2$ is true.

E ($\phi 1\ U\ \phi 2$): From a given state, There is a path where $\phi 1$ holds until $\phi 2$ is true.

Now we will illustrate equivalence of some basic CTL formulas

i) $\neg AF\varphi \equiv EG\neg\varphi$

$\neg AF\varphi$: From a given state the following does not hold "In all paths in future ϕ is true"

$EG\neg\varphi$ =From a given state the following holds "There is a path where globally ϕ is not true"

A and G corresponds to universal quantifier (in predicate logic) whereas E and F corresponds existential quantifier (in predicate logic). So Se Morgan's rules holds for A and E and also F and G. Also we have $AF\varphi \equiv \neg EG\neg\varphi$, $EF\varphi \equiv \neg AG\neg\varphi$ etc.

Note: ***A*** is dual of ***E*** and ***G*** is dual of ***F***.

Illustration by an example: Figure 2 illustrates three cases where in state s0, $EG\neg\varphi$ holds because there exists a path from s0, where in future there is no state which contains φ. It may be noted from Figure 2 that $AF\varphi$ is not true in s0 because there are paths from s0 where states does not contain ϕ in future; so $\neg AF\varphi$ holds in s0. So, we have $\neg AF\varphi \equiv EG\neg\varphi$ in the example.

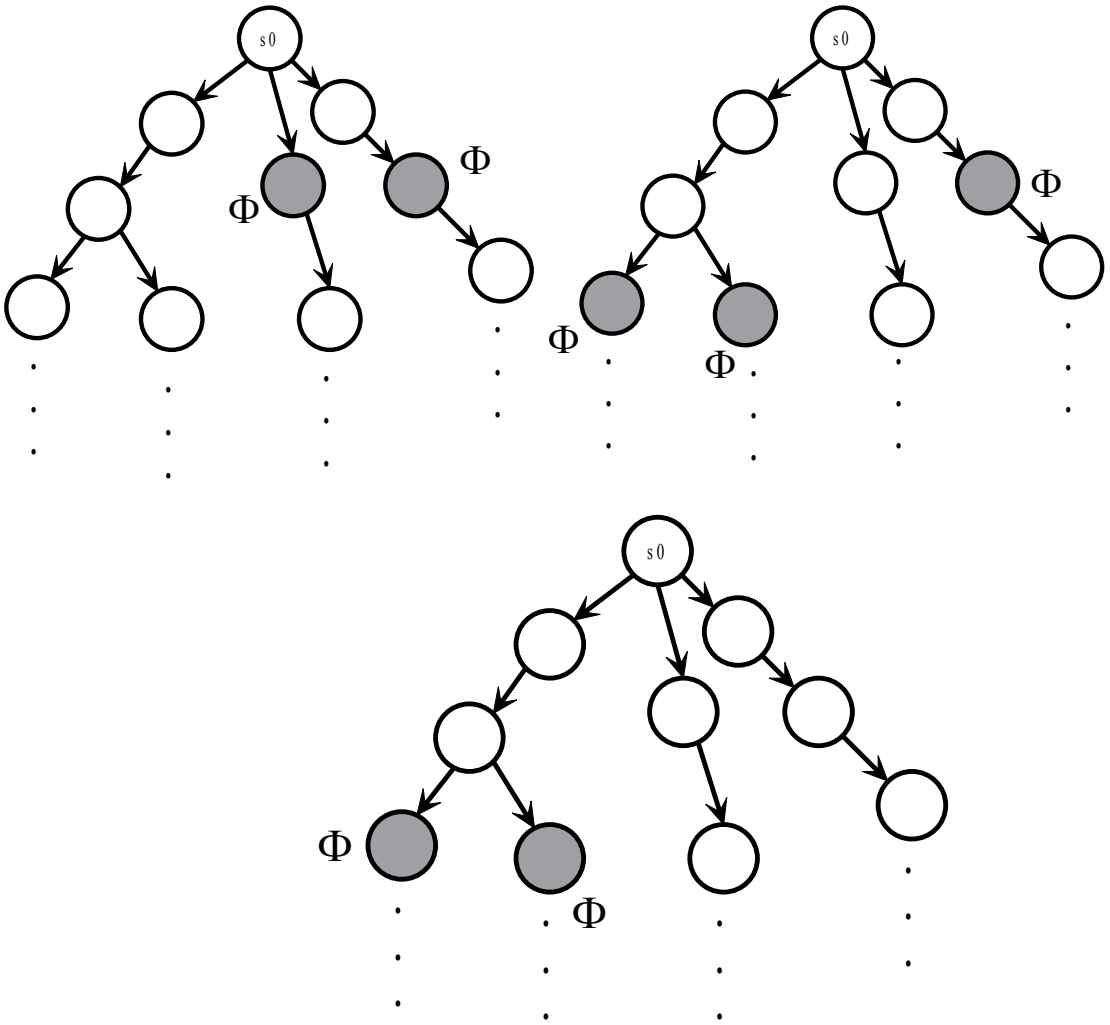


Figure 2. Three cases where in state s0, $EG\neg\varphi$ hold

ii) $\neg EF\varphi \equiv AG\neg\varphi$

$\neg EF\varphi$: From a given state the following does not hold "There is a path where in future ϕ is true"

$AG\neg\varphi$ = From a given state the following holds "In all paths globally ϕ is not true"

This holds by De Morgan's laws.

Illustration by an example: Figure 3 illustrates a case where in state s0, $AG\neg\varphi$ holds, because in all paths from s0 no state satisfies φ. It may be noted from Figure 3 that $\neg EF\varphi$ is not true in s0 because there is no path from s0 where states contain ϕ in future. So, we have $\neg EF\varphi \equiv AG\neg\varphi$ the example. We may also write $EF\varphi \equiv \neg AG\neg\varphi$.

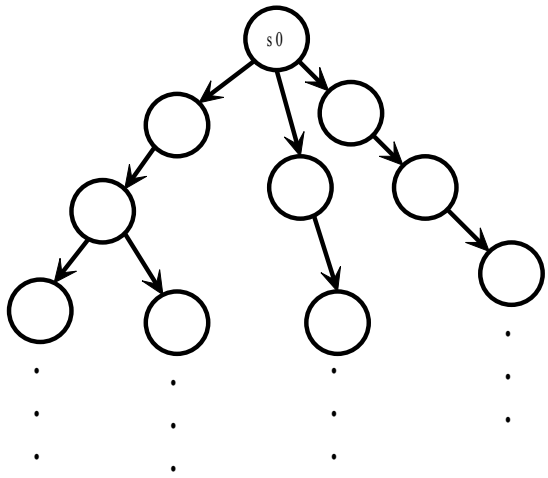


Figure 3. $AG\neg\varphi$ holds in s0

iii) $\neg AX\varphi \equiv EX\neg\varphi$

$\neg AX\varphi$: From a given state the following does not hold "In all paths next state satisfies ϕ".

$EX\neg\varphi \equiv$ From a given state the following holds "There exist a path where in next state ϕ is not true.

Illustration by an example: Figure 4 illustrates two cases where in state s0, $\neg AX\varphi$ holds, because there is a path from s0 where next state do not satisfy φ. It may be noted from Figure 4 that $EX\neg\varphi$ is true in s0 because there is a path from s0 where next state do not satisfy φ. So, we have $\neg AX\varphi \equiv EX\neg\varphi$ the example. We may also write $AX\varphi \equiv \neg EX\neg\varphi$.

Note: *X* is dual of itself.

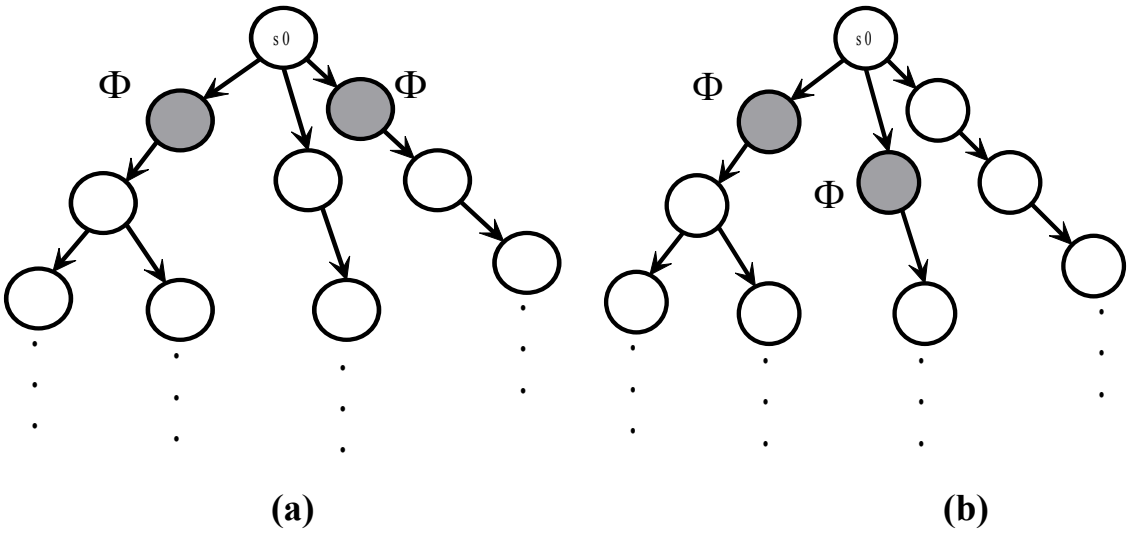


Figure 4. Two cases where in state s0, $\neg AX\varphi$ hold

iv) $AF\varphi \equiv A[TU\varphi]$

$AF\varphi$ **:** From a given state the following holds "In all paths there is a state where in future ϕ is true.

$A[TU\varphi]$**:** From a given state the following holds "In all paths T (true) until ϕ holds".

Illustration by an example: Figure 5 illustrates a case where in state s0, $AF\varphi$ holds. We know $\varphi 1 U \varphi 2$ implies that $\phi 1$ is true until $\phi 2$ becomes true. Now, let $\phi 1$ be True (T). Since "T" is true in every state we can write $TU\varphi 2$; T (i.e., every state) holds until $\varphi 2$ becomes true, which is $F\varphi$. So, it may be noted from Figure 5 that $A[TU\varphi]$ is true in s0 because all paths from s0 satisfy $F\varphi$. So, we have $AF\varphi \equiv A[TU\varphi]$.

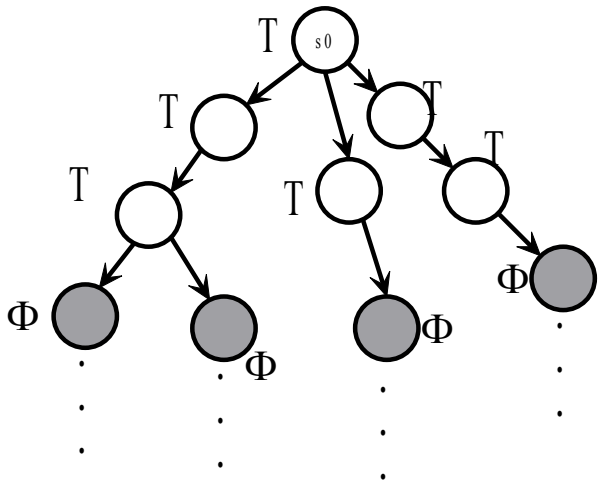


Figure 5. In state s0, $AF\varphi$ and $A[TU\varphi]$ hold

v) $EF\varphi \equiv E[TU\varphi]$

The equivalence can be shown similar to the one discussed in point (iv); this is illustrated in Figure 6

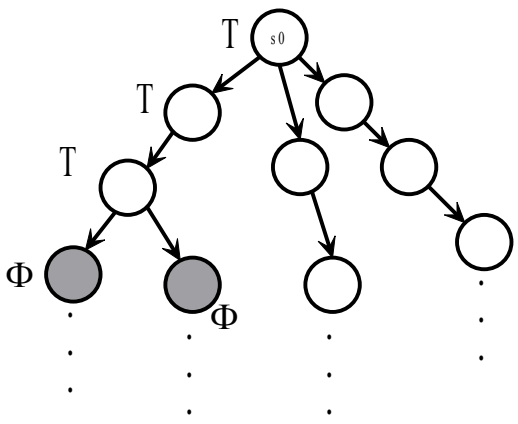


Figure 5. In state s0, $EF\varphi$ and $E[TU\varphi]$ hold

Now, we can have the following theorem

Theorem 1: The set of operators $\bot, \neg, \wedge$ together with *AU, EU* and *EX* are adequate for CTL. Any CTL formula can be transformed into a semantically equivalent CTL formula which uses only those logical connectives.

Proof: From the equivalence of five CTL formulas discussed above we note that

- AX can be written in terms of EX ; equivalence (iii)
- AF can be written in terms of AU ; equivalence (iv)
- EF can be written in terms of EU ; equivalence (v)
- AG can be written in terms of EF ; equivalence (ii). Also, EF can be written in terms of EU
- EG can be written in terms of AF ; equivalence (i). Also, AF can be written in terms of *AU,*

Similarly we can have the following as adequate sets together with $\bot, \neg, \wedge$.

$$\boldsymbol{AF, \quad EU, \quad EX}$$
$$\boldsymbol{EG, \quad EU, \quad EX}$$
$$\boldsymbol{AG, \quad AU, \quad AX}$$

To show the above as complete sets we need to repeat the steps as done for Theorem 1. All these steps are similar and simple, except to show that AU can be represented in terms of EU and vice-versa.

Now, we will show that $A[\varphi 1 U \varphi 2] \equiv \neg(E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)] \vee EG\neg\varphi 2)$, which illustrates how AU can be represented in terms of EU.

We have,

$\neg(E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)] \vee EG\neg\varphi 2)$

$\equiv \neg E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)] \wedge \neg EG\neg\varphi 2$

$\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)$: A path in which first in sequence come some states that contains $\neg\varphi 2$ (may contain $\varphi 1$) and next states contain $\neg\varphi 1 \wedge \neg\varphi 2$ (do not contain $\varphi 1$). In other words, a path in which first in sequence come some states that contain $\varphi 1$ (not $\varphi 2$) and next states do not contain $\varphi 1$ or $\varphi 2$.

$E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)]$**:** There exists a state from which there is a path in which first in sequence come some states that does not contains $\varphi 2$ and next states contain $\neg\varphi 1 \wedge \neg\varphi 2$.

$\neg E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)]$

$\equiv A[\neg(\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2))]$**:**

There exists a state from which in all paths the following holds—"first in sequence come some states that contains $\varphi 2$ (and may be $\varphi 1$) and next states contain $\neg\varphi 1 \wedge \neg\varphi 2$"

$\neg EG\neg\varphi 2 \equiv AF\varphi 2$: From a state, in all paths in a future state $\varphi 2$ holds.

So, $A[\varphi 1 U \varphi 2]$ is made of two parts $\neg E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)]$ and $\neg EG\neg\varphi 2 \equiv AF\varphi 2$ which implies that—There exists a state from which in all paths the following holds "first in sequence come some states that contains $\varphi 2$ (and may be $\varphi 1$) and in future $\varphi 2$ holds (before we have states where neither $\varphi 1$ nor $\varphi 2$ holds). It may be noted that this statement is equivalent to--There exists a state from which in all paths the following holds--$\varphi 1$ holds until $\varphi 2$ holds.

So, $\neg(E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)] \vee EG\neg\varphi 2) \equiv A[\varphi 1 U \varphi 2]$. Figure 6 illustrates the proof concept.

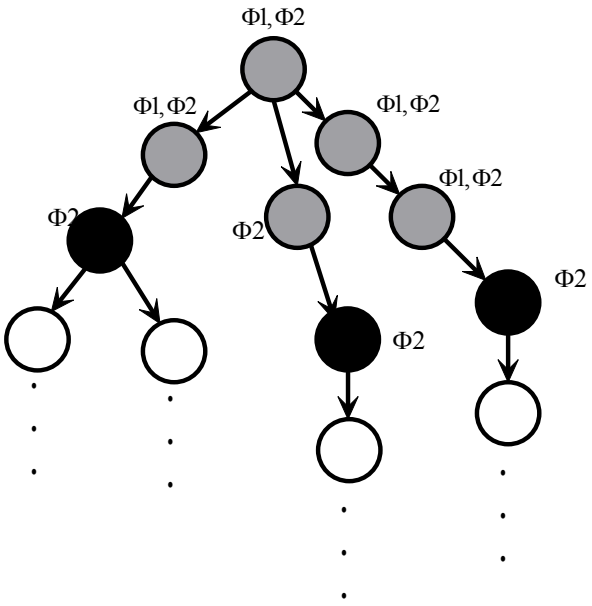


Figure 6. Proof concept of $A[\varphi 1 U \varphi 2] \equiv \neg(E[\neg\varphi 2 U(\neg\varphi 1 \wedge \neg\varphi 2)] \vee EG\neg\varphi 2)$

3. Recurrence Equivalence of CTL formulas

The basic idea of recurrence based equivalence is based on the following idea:

A given CTL formula φ, which is satisfied in state s0, it is broken into two parts

- In state s0 we put some CTL formula which is true only in that state
- The second part comprises CTL formula which is true in one or all next states of s0.

i). $\boldsymbol{AG\phi \equiv \phi \wedge AXAG\phi}$

$\boldsymbol{AG\phi \equiv}$ From a given state in all paths globally $\boldsymbol{\phi}$ is true.

$\boldsymbol{\phi \wedge AXAG\phi \equiv}$ Present state satisfies $\boldsymbol{\phi}$ and for all next states, in all paths $\boldsymbol{\phi}$ is globally true. Figure 7 illustrates the concept.

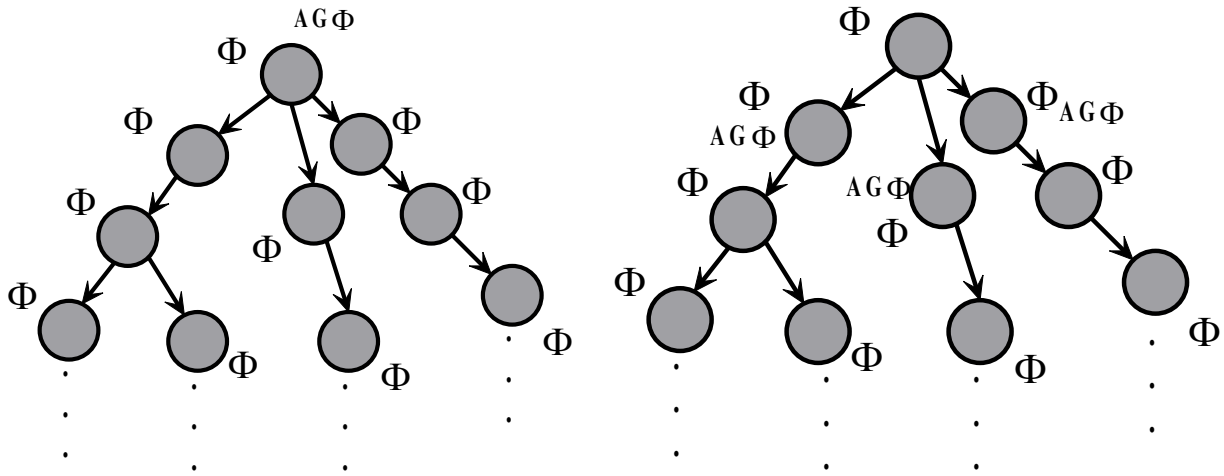


Figure 7. Illustration of $\boldsymbol{AG\phi \equiv \phi \wedge AXAG\phi}$

ii). $\boldsymbol{EG\phi \equiv \phi \wedge EXEG\phi}$

$\boldsymbol{EG\phi \equiv}$ From a given state there exist a path in which globally $\boldsymbol{\phi}$ is true.

$\boldsymbol{\phi \wedge EXEG\phi \equiv}$ Present state satisfies $\boldsymbol{\phi}$ and there is some next state, from which there is some path in which $\boldsymbol{\phi}$ is globally true. Figure 8 illustrates the concept.

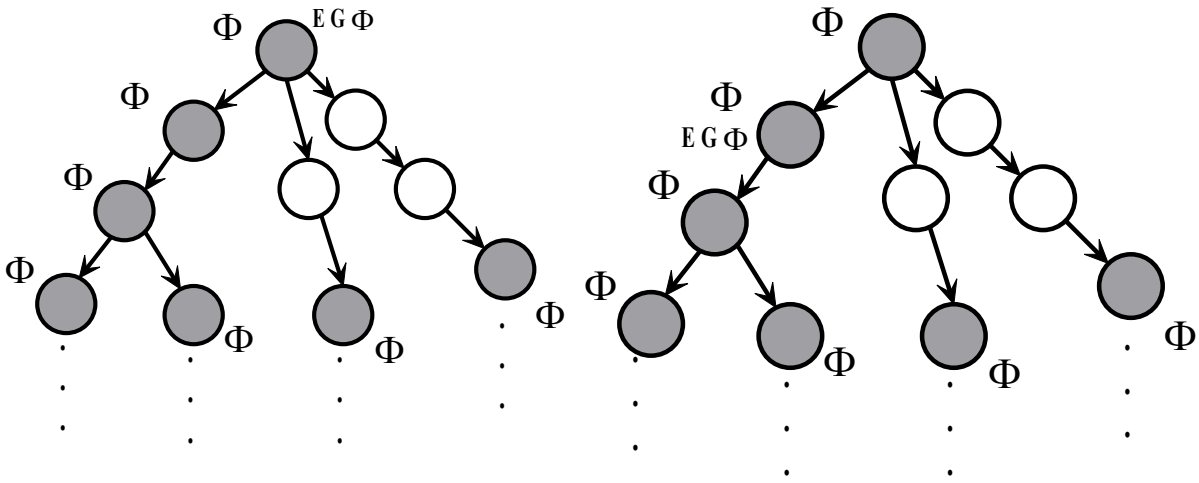


Figure 8. Illustration of $EG\phi \equiv \phi \wedge EXEG\phi$

iii). $AF\phi \equiv \phi \vee AXAF\phi$

$AF\phi \equiv$ From a given state in all paths there in future where ϕ is true.

$\phi \vee AXAF\phi \equiv$ Present state satisfies ϕ or for all next states, in all paths from next state there is a state in future where ϕ is true. Figure 9 illustrates the concept.

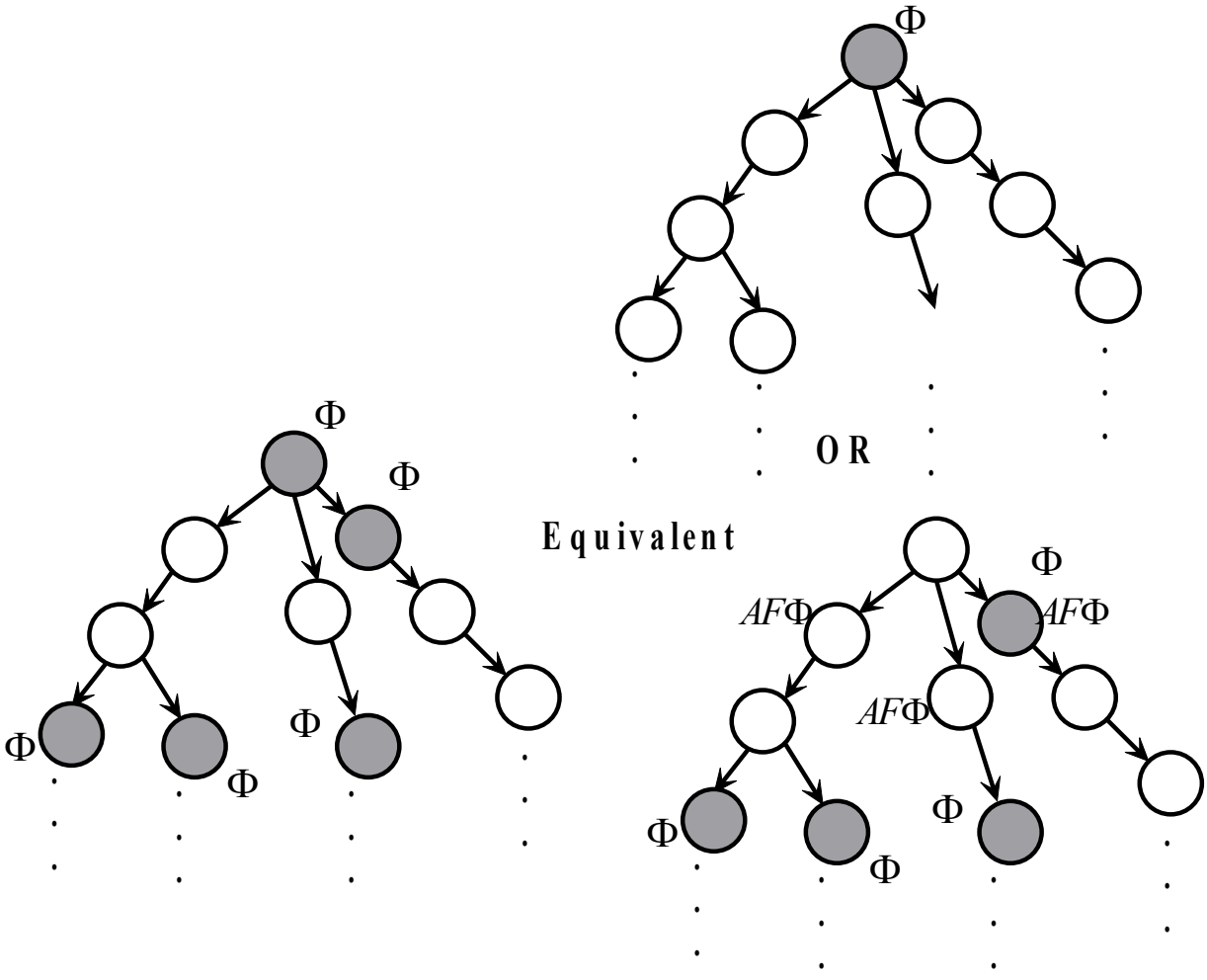


Figure 9. Illustration of $AF\phi \equiv \phi \vee AXAF\phi$

iv). $\boldsymbol{EF\phi \equiv \phi \vee EXEF\phi}$

$\boldsymbol{EF\phi}$ ≡ From a given state there exists a path in which in future $\boldsymbol{\phi}$ is true.

$\boldsymbol{\phi \vee EXEF\phi}$ ≡ Present state satisfies $\boldsymbol{\phi}$ or there exists some next state from which in some path in future $\boldsymbol{\phi}$ is true. Figure 10 illustrates the concept.

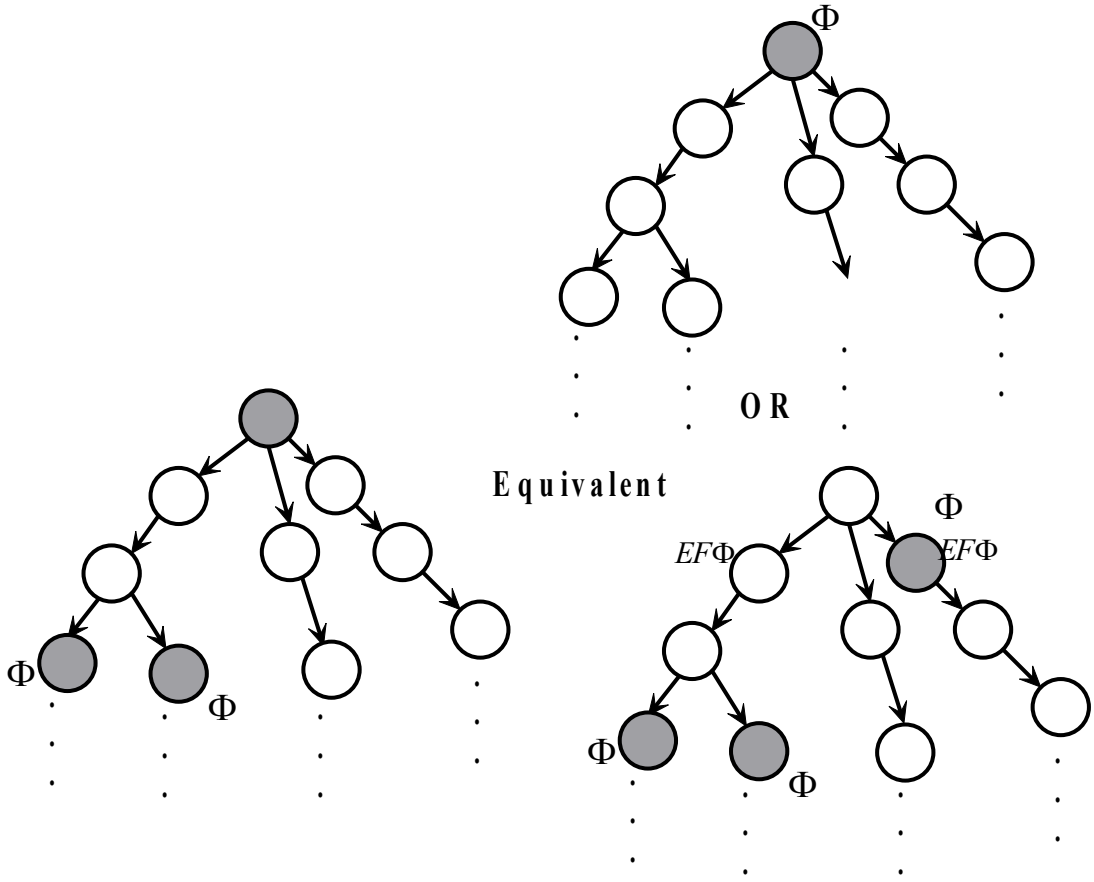


Figure 10. Illustration of $\boldsymbol{EF\phi \equiv \phi \vee EXEF\phi}$

v). $\boldsymbol{A[\phi U\psi] \equiv \psi \vee [\phi \wedge AXA[\phi U\psi]]}$

$\boldsymbol{A[\phi U\psi]}$ ≡ In all paths from current state $\boldsymbol{\phi}$ is true until $\boldsymbol{\psi}$ becomes true.

$A[\phi U\psi] \equiv \psi \vee [\phi \wedge AXA[\phi U\psi]]$ Present state satisfies $\boldsymbol{\psi}$ or present state satisfies $\boldsymbol{\phi}$ and from all next states in all paths $\boldsymbol{\phi}$ holds until $\boldsymbol{\psi}$ becomes true. Figure 11 illustrates the concept.

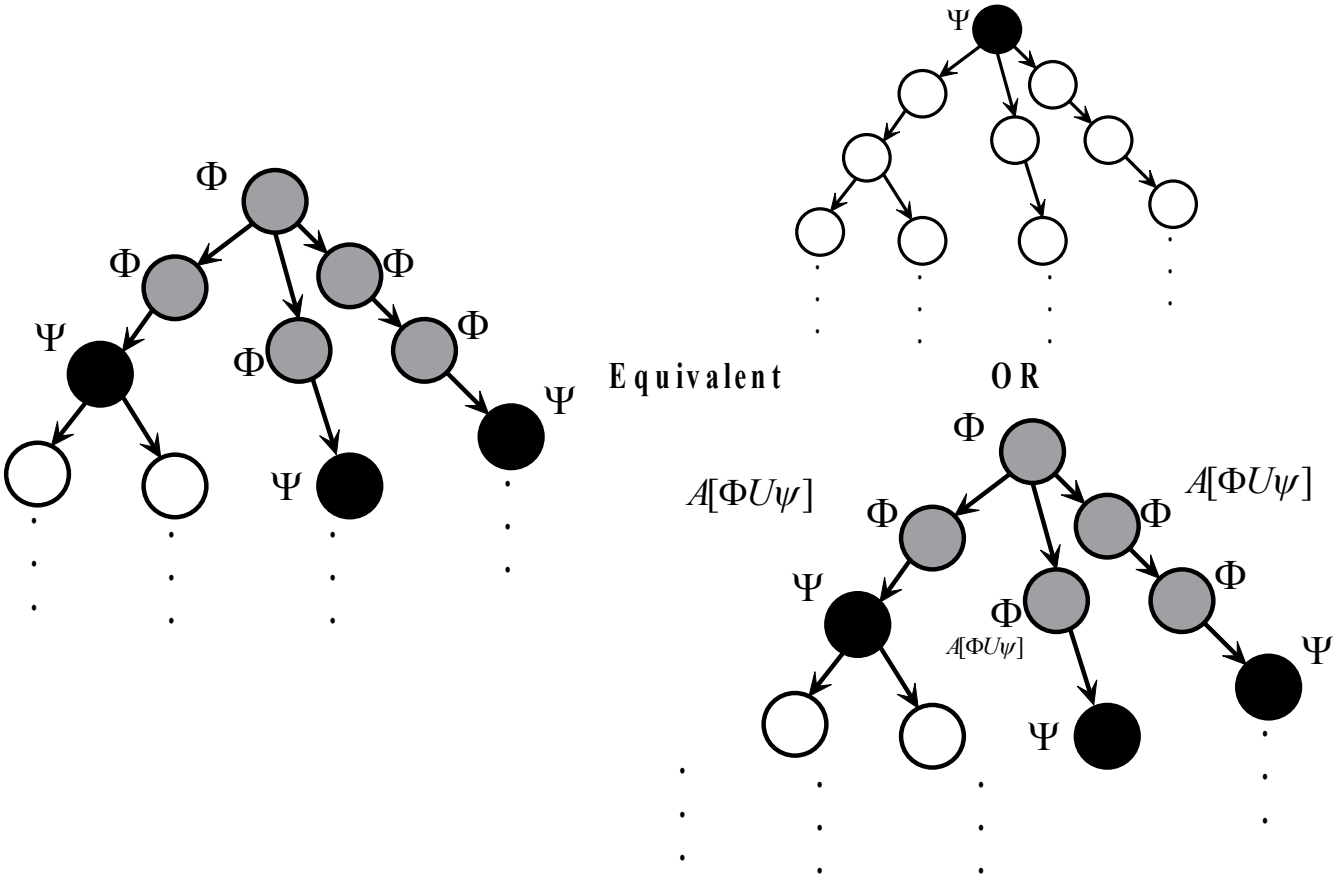


Figure 11. Illustration of $\boldsymbol{A[\phi U \psi] \equiv \psi \vee [\phi \wedge AXA[\phi U \psi]]}$

vi). $\boldsymbol{E[\phi U \psi] \equiv \psi \vee [\phi \wedge EXE[\phi U \psi]]}$

$\boldsymbol{E[\phi U \psi] \equiv}$ There exists a part from a given state where $\boldsymbol{\phi}$ is true until $\boldsymbol{\psi}$ became true.

$E[\phi U \psi] \equiv \psi \vee [\phi \wedge EXE[\phi U \psi]]$**:** In present state $\boldsymbol{\psi}$ is true **or** present state satisfies $\boldsymbol{\psi}$ and there exist some next state from where there exists some path where $\boldsymbol{\phi}$ holds until $\boldsymbol{\psi}$ becomes true. This concept is similar to the one illustrated in equivalence (iv) where "for all paths" is replaced by "there exists a path".

4. **Brief overview of CTL versus other temporal logic LTC and CTL***

Till now, we have mainly discussed about one class of temporal logic i.e., CTL. As discussed in lecture 1 of this module, LTL and CLT* are other important classes of temporal logic. In this course we will not discuss in detail regarding the LTL and CLT*; the details can be found in [1]. However, we will briefly introduce LTL and CTL* and compare their expressive power with CTL.

4.1 Linear time temporal logic (LTL)

Linear-time temporal logic (LTL) has the following syntax:

$\Phi ::= \quad \bot \mid \top \mid P \mid (\neg\varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \rightarrow \varphi) \mid X\varphi \mid F\varphi \mid G\varphi \mid \varphi \, U \, \varphi$ Where p is any propositional atom.

Unlike CTL which is evaluated on states, an LTL formula is evaluated on a path, or a set of paths. A path is defined as $\pi = s_1 \rightarrow s_2 \cdots$ and we write $\pi^i = s_i \rightarrow s_{i+1} \rightarrow s_{i+2} \cdots$ for suffix paths starting at s_i. A set of paths satisfies Φ if every path in the set satisfies Φ.

Like CTL, the semantics of LTL is defined over a model M: $\langle S, \rightarrow, L \rangle$. Let a path be defined as $\pi = s_1 \rightarrow s_2 \cdots$ in M. Whether π satisfies an LTL formula is defined by the satisfaction relation $\models$ as follows.

- $\pi \models \top$ and $\pi \not\models \bot$; "truth is true and false is false in π"
- $\pi \models p$ iff $p \in L(s_1)$; atomic proposition p is satisfied in π if label of s_1 has p.
- $\pi \models \neg \Phi$ iff $\pi \not\models \Phi$; $\neg \Phi$ is satisfied in π if π does not satisfy $\neg \phi$
- $\pi \models \Phi_1 \wedge \Phi_2$ iff $\pi \models \Phi_1$ and $\pi \models \Phi_2$; $\Phi_1 \wedge \Phi_2$ is satisfied in π if both Φ_1 and Φ_2 are satisfied in π.
- $\pi \models \Phi_1 \vee \Phi_2$ iff $\pi \models \Phi_1$ or $\pi \models \Phi_2$; $\Phi_1 \vee \Phi_2$ is satisfied in π if Φ_1 or Φ_2 is satisfied in π.
- $\pi \models \Phi_1 \rightarrow \Phi_2$ iff $\pi \models \Phi_2$ whenever $\pi \models \Phi_1$.
- $\pi \models X\Phi$ iff $\pi^2 \models \Phi$; $X\Phi$ holds in π if s_2 (i.e., next of s_1) satisfies Φ.
- $\pi \models G\Phi$ iff, for all $i \geq 1$, $\pi^i \models \Phi$; $G\Phi$ holds in π if all its states satisfies Φ.
- $\pi \models F\Phi$ iff there is some i ≥ 1 such that $\pi^i \models \Phi$; $F\Phi$ holds in π if some future state of π satisfies Φ.
- $\pi \models \Phi$ U ψ iff there is some i ≥ 1 such that $\pi^j \models \psi$ and for all $j = 1, \ldots, (i-1)$ we have $\pi^j \models \Phi$; Φ U ψ holds in π if from some state s_1, Φ U ψ holds.

4.2 The Expressive Powers of LTL and CTL

CTL allows explicit quantification over paths from a state (*A* for all paths and E for at least one path); in this respect it is more expressive than LTL because in LTL we cannot have explicit quantification over paths. However, CTL does not allow selection of a range of paths which is possible in LTL; in that respect, LTL is more expressive. For example, if we want to write

"All paths which have a *p* along them also have a *q* along them". We can write this in LTL as $Fp \rightarrow Fq$.

Let us try to write it in CTL. The formula $AFp \rightarrow AFq$ is close to the LTL formula written above. $AFp \rightarrow AFq$ means something different: it says "if all paths have a *p* along them in future, then all paths have a *q* along them in future". Let us try $AG(p \rightarrow AF\ q)$, which says "in all paths from a given state globally the following holds--if *p* holds then in all paths in future state *q* holds. In other words, every way of extending every path to a *p* eventually meets a *q*, but that is still not capturing the meaning of $Fp \rightarrow Fq$ .It may be noted that it is not possible to write $Fp \rightarrow Fq$ in CTL because every *F* has an associated *A* or *E*. CTL* eliminates this issue, where the CTL constraint that every temporal operator *(X, U, F, G)* has to be associated with a unique path quantifier *(A, E)* is eliminated.

4.3 The Expressive Powers of LTL and CTL*

CTL* is a logic which combines the expressive powers of LTL and CTL, by dropping the CTL constraint. Some examples of CTL* formulas are as follows:

- *E [X*$\phi \vee G\ \psi$ *]:* there is a path, either ϕ is true in next state, or ψ is true in all states.
- *A[Fp* $\wedge$ *Fq]:* along all paths, both *p* and *q* are true in future.
- *E [F*$\phi \vee (\phi 1\ \mathrm{U}\ \phi 2)$*]:* there is a path, either ϕ is true in future or $\phi 1$ is true until $\phi 2$.

CTL* involves two classes of formulas:

- **state formulas**:

  ϕ :: = T | *p* | (ϕ) | ($\phi \wedge \phi$) | A[α] | E[α]

  Where *p* is any atomic formula and α any path formula;
- **path formulas**:

α ::= ϕ | (α) | (α ∧ α) | (α U α) | (G α) | (F α) | (X α) where ϕ is any state formula.

4.4 LTL and CTL as subsets of CTL*

LTL does not include the concept of all paths or at least one path(*A* or *E)*, rather the semantic viewpoint of LTL is that we consider All paths. In other words, an LTL formula α is equivalent to the CTL* formula A[α]. Thus, LTL can be viewed as a subset of CTL*. CTL is also a subset of CTL*, since it is the subset of CTL* where the path formulas are restricted to **α**:= (ϕ Uϕ) | (Gϕ) | (F ϕ) | (X ϕ). Figure 12 illustrates the set-subset relationship of expressive power of LTL, CTL and CTL*.

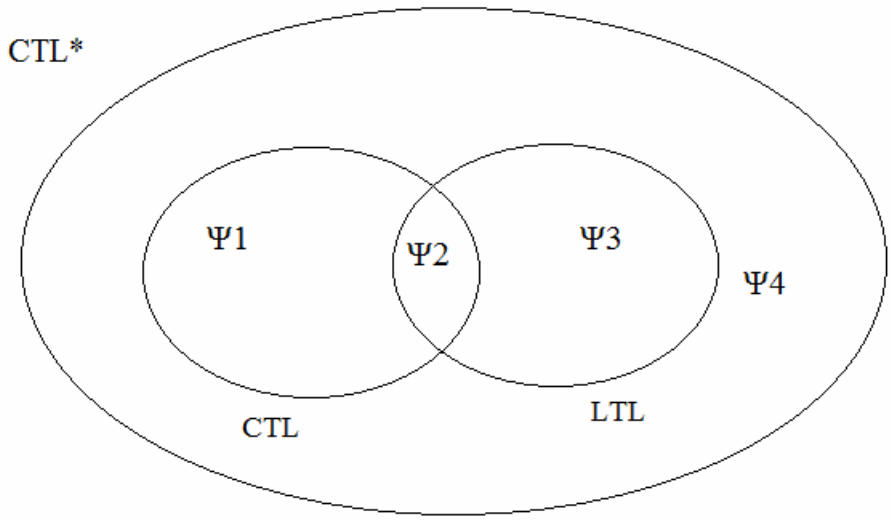


Figure 12. The expressive power of LTL, CTL and CTL*

5. Conclusions

In this module we have discussed about the specification language CTL in details and looked into some other temporal logic like LTL, CTL*, etc. By looking into their syntax, we can say that their expressive power is different. It is quite obvious that if we go for more expressive logic, the reasoning about the logic will be more expensive. In next module we will discuss how the specification language CTL is used for verification. In other words, in the next module we will discuss, given a CTL formula and a model, how automatically it can be verified if a given state satisfies the formula.

References

[1]. Michael Huth and Mark Ryan, “Logic in Computer Science: Modelling and Reasoning about Systems”, 2nd edition, Cambridge University Press, New York, NY, USA.

Question and Answers

1. **Question**: Give an example of a model to illustrate the following LTL formulas

   $\pi \models X\psi$, $\pi \models G\Phi$, $\pi \models \Phi \mathrm{U}\ \psi$.

Answer:

In the figure given below $\pi 3 \models X\psi$, $\pi 1 \models G\Phi$ and $\pi 1 \models \Phi \mathrm{U}\ \psi$.

